AEROSPACE ENGINEERING · STUDENT PROJECT

Lambert Studio

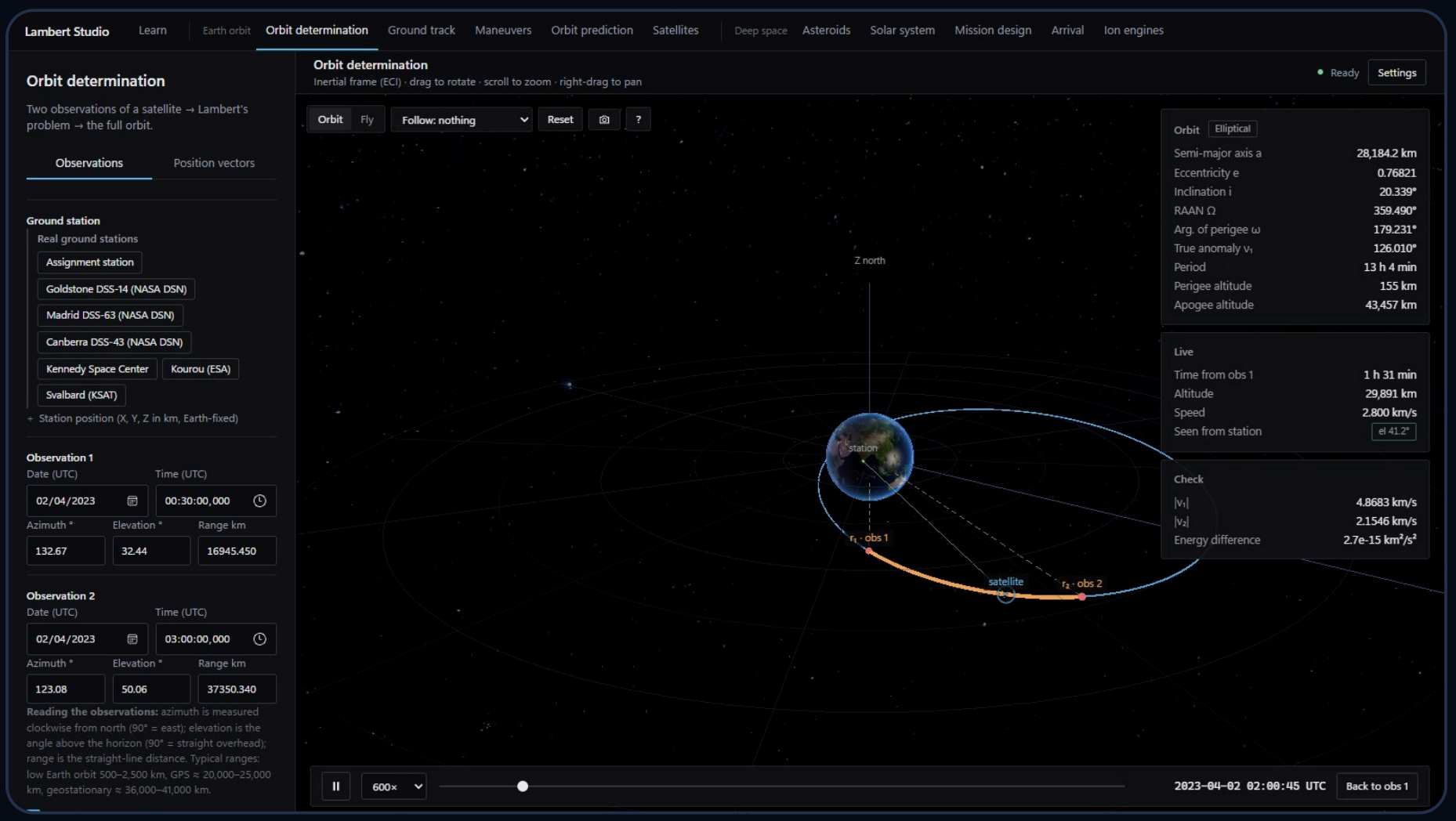
I turned a university assignment on orbit determination into an **interactive mission-design studio** that rediscovers the routes of Voyager, Cassini and Galileo from physics alone.



THE STARTING POINT

Two sightings of a satellite. Find its orbit.

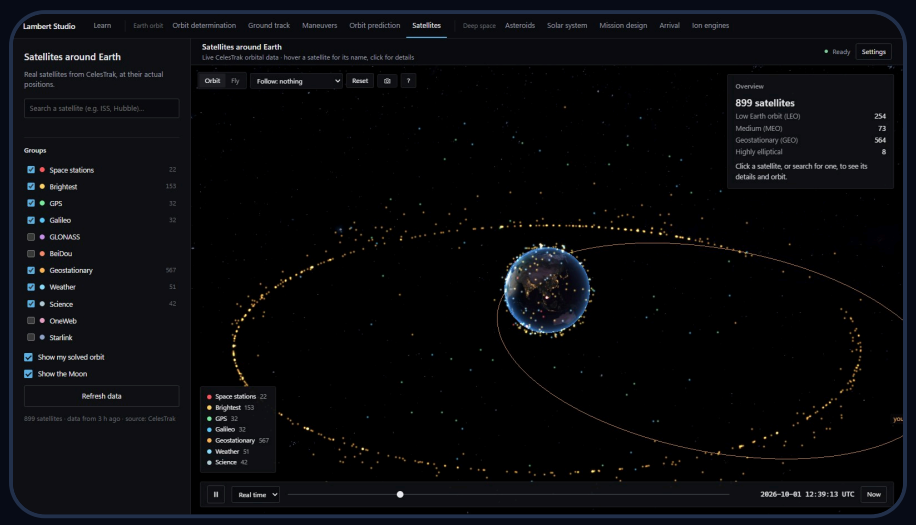
A ground station measures azimuth, elevation and range twice, 2.5 hours apart. Solving **Lambert's problem** (the maths behind every interplanetary mission) turns those six numbers into the complete orbit.



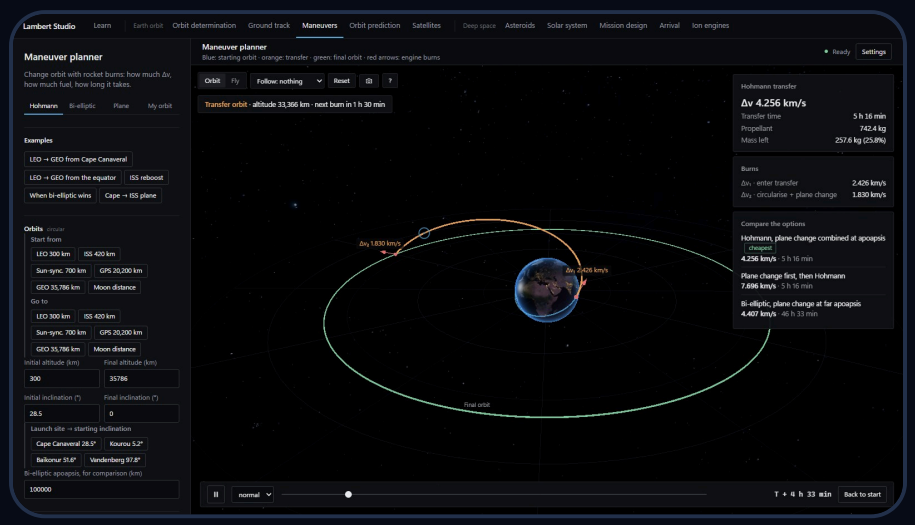
a = 28,184 km · e = 0.768 · i = 20.34° · period 13 h 04 min

WHAT IT DOES

Eleven modules, one desktop app



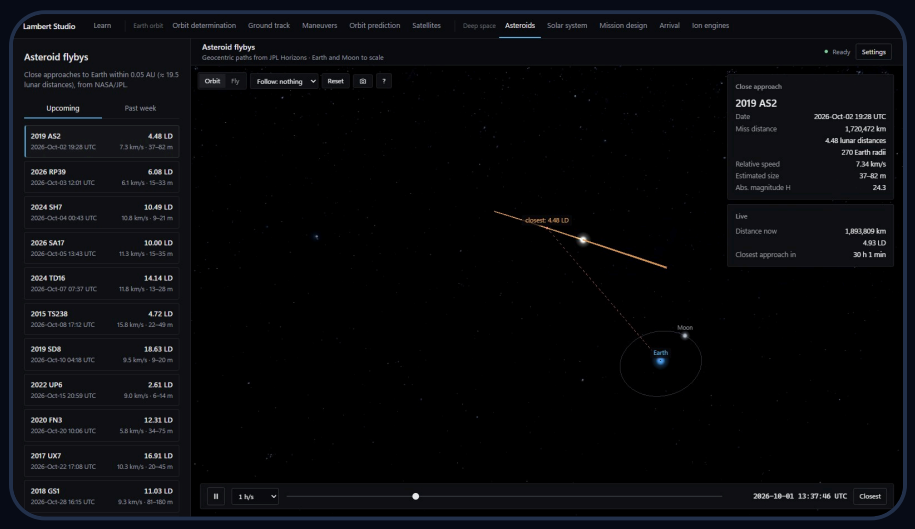
Real satellites (CelesTrak)



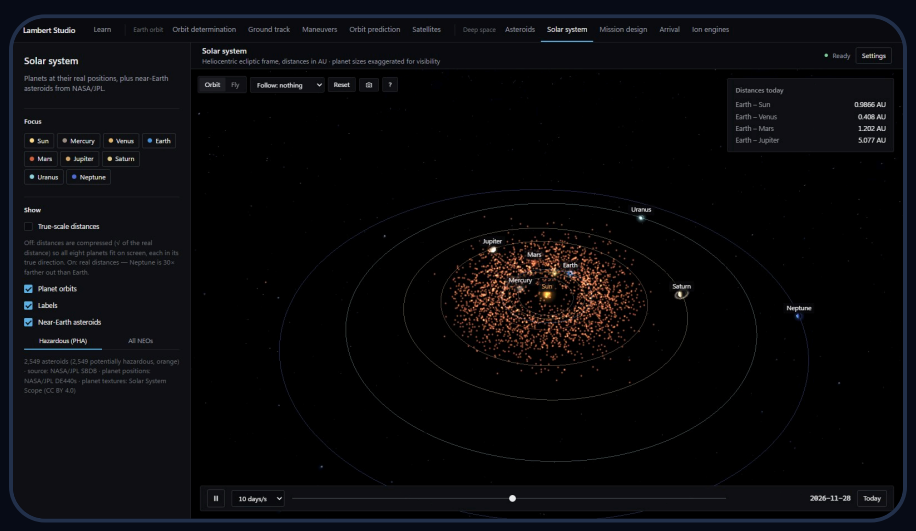
Manoeuvre planner



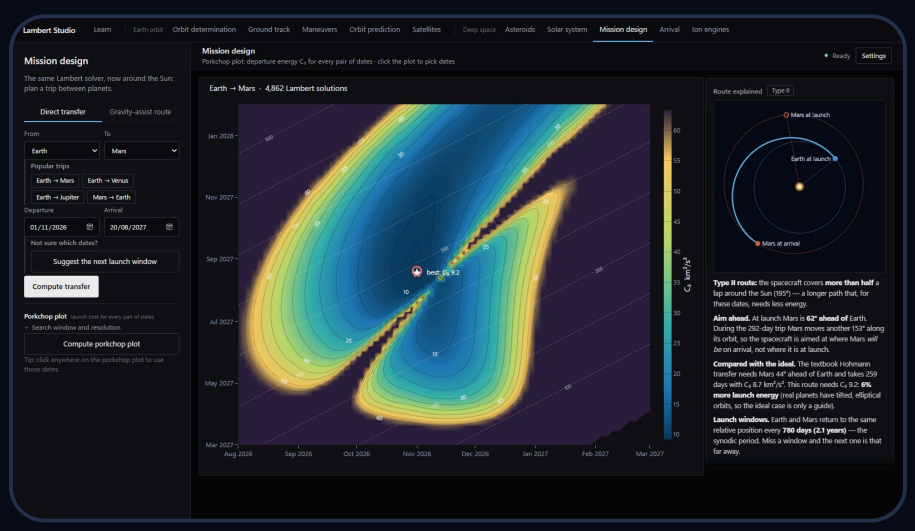
Full-force orbit prediction



Asteroid flybys (NASA/JPL)



Solar system (JPL DE440s)



Porkchop & launch windows

GRAVITY ASSISTS

Let the planets do the work

Routes with up to four flybys, including repeat visits to the same planet (multi-revolution Lambert), optimised over thousands of launch dates and flight times, then **explained step by step**.



Cassini's launch period: flying direct to Saturn costs **20.7 km/s**; the flyby route needs **7.8 km/s**.

MISSION PLAYBACK

+10.48 km/s

Voyager at Jupiter, May 1979. No fuel.



A full-screen, cinematic replay of any mission: the camera swings to each planet, time slows at every flyby, and live telemetry shows the spacecraft passing solar escape speed.

VALIDATED

It rediscovered history

The optimiser is never told the real dates, only the planets and a launch period.

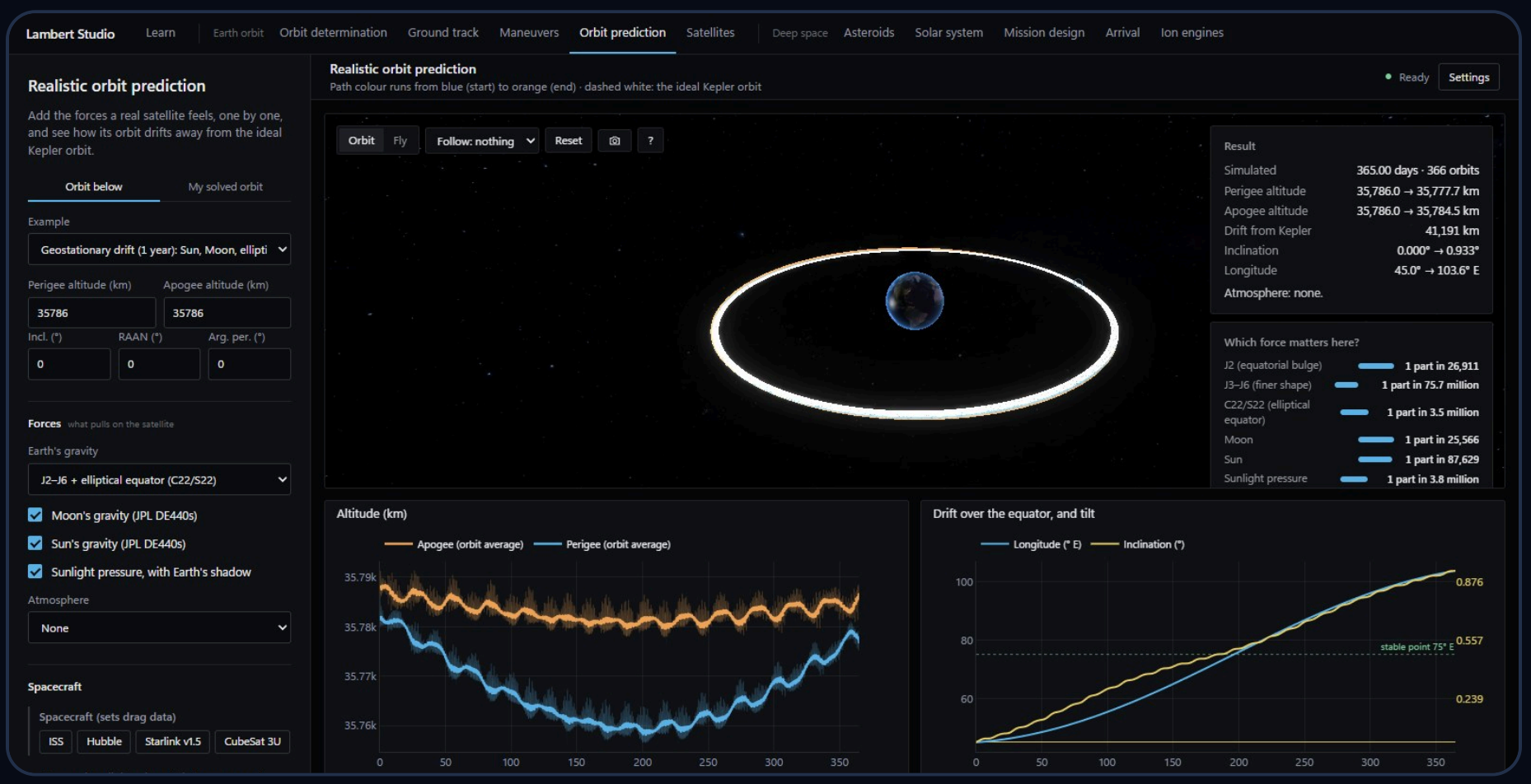
Mariner 10 · Venus flyby	6 Feb 1974	1.6 days off
Voyager · launch	2 Sep 1977	2.2 days off
Galileo · Venus flyby	21 Feb 1990	11 days off
Cassini · launch	16 Oct 1997	1.4 days off
Cassini · Earth flyby	17 Aug 1999	4 hours off
Hohmann LEO → GEO	3.8926 km/s	textbook 3.893

136 automated validation checks · **58** pytest tests · browser-driven UI tests

THEN I MADE IT MORE ACCURATE

Real forces. Real gravity. Real corrections.

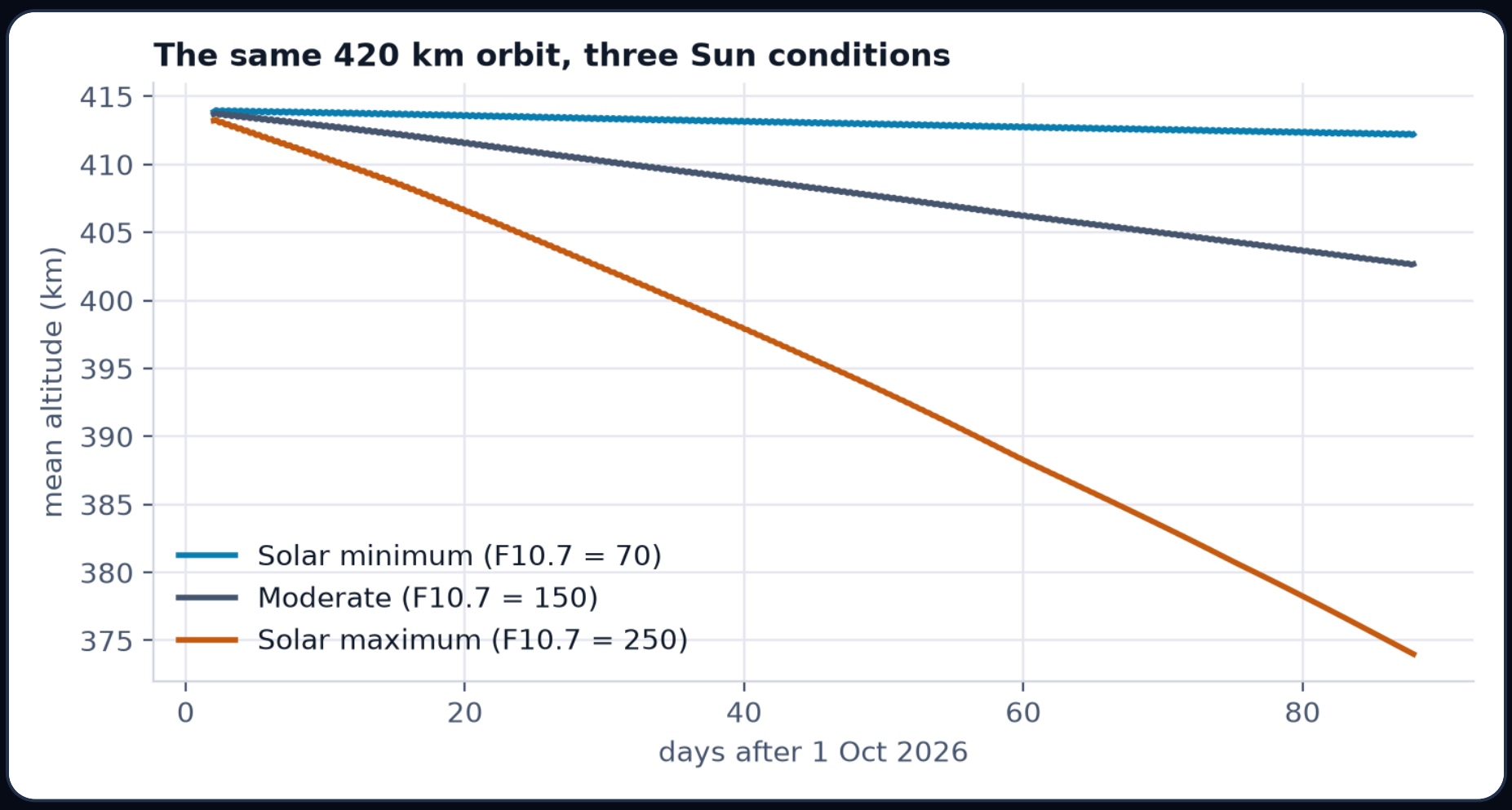
Earth's gravity to degree 6, the Moon and Sun from JPL, sunlight pressure with Earth's shadow and the NRLMSIS atmosphere. Routes are re-flown with every planet pulling at once.



A geostationary satellite tilts 0.95° in a year, inside the 0.75–0.95° real operators see.

HOW ANSWERS CHANGE WITH TIME

Same orbit, 23× faster decay at solar maximum



Mars launch energy varies almost 2× between windows. Mars is 3 to 22 light-minutes away. Voyager's simple-model route would hit Jupiter until **4 m/s** of corrections fix it. **A simulation answers for a date, not forever.**

HOW IT WAS BUILT

Nine phases, each verified before the next

01 Rebuilt the Lambert solver and checked it against textbook examples

02 Desktop app: Python physics + local API + Three.js/WebGL interface

03 Live data: CelesTrak, NASA/JPL Horizons, SBDB, DE440s ephemeris

04 Manoeuvres, and a J2 + drag integrator (SciPy DOP853)

05 Gravity assists: batch + multi-revolution Lambert, beam search

06 Mission playback, free camera, NASA-sourced input suggestions

07 pytest suite, Windows .exe (PyInstaller), course hand-in

08 Learn tab, moon tours, ion engines, orbit uncertainty

09 High-fidelity forces + full solar-system route check

STACK & TAKEAWAYS

≈20,000 lines. One lesson: verify everything.

Python | NumPy | SciPy | Skyfield | NRLMSIS | sgp4 | Three.js | Plotly | pytest | PyInstaller | git

Screenshots can look right and still be wrong; a wider search turned Cassini from worse-than-direct into the historical route; and knowing where the maths breaks (resonant orbits) matters as much as the maths.

Built with **Claude Code** as an AI pair-programming assistant, with every step directed, run and checked by me.